



A hybrid Framework for plant leaf disease detection and classification using convolutional neural networks and vision transformer

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Abstract

Recently, scientists have widely utilized Artificial Intelligence (AI) approaches in intelligent agriculture to increase the productivity of the agriculture sector and overcome a wide range of problems. Detection and classification of plant diseases is a challenging problem due to the vast numbers of plants worldwide and the numerous diseases that negatively affect the production of different crops. Early detection and accurate classification of plant diseases is the goal of any AI-based system. This paper proposes a hybrid framework to improve classification accuracy for plant leaf diseases significantly. This proposed model leverages the strength of Convolutional Neural Networks (CNNs) and Vision Transformers (ViT), where an ensemble model, which consists of the well-known CNN architectures VGG16, Inception-V3, and DenseNet20, is used to extract robust global features. Then, a ViT model is used to extract local features to detect plant diseases precisely. The performance proposed model is evaluated using two publicly available datasets (Apple and Corn). Each dataset consists of four classes. The proposed hybrid model successfully detects and classifies multi-class plant leaf diseases and outperforms similar recently published methods, where the proposed hybrid model achieved an accuracy rate of 99.24% and 98% for the apple and corn datasets.

Keywords Farming · Plant leaf disease classification · Hybrid model · Deep learning · Convolutional neural networks (CNNs) · Feature concatenation · Vision transformer (ViT)

Abbreviations

AI	Artificial Intelligence
CNNs	Convolutional Neural Networks
ViT	Vision Transformer
ML	Machine Learning
DL	Deep Learning

SVM	Support Vector Machine
ANNs	Artificial Neural Networks
NB	Naive Bayes
MLP	Multilayer Perception
MHA	Multi-Head self-Attention
GP	Global Pooling
FFN	Feed Forward Network
NL	Normalization Layer
LR	Learning Rate
AUC	Area Under Curve
ROC	Receiver Operating Characteristic
CM	Confusion Matrix

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Introduction

Plant diseases pose an enormous risk to the world's food supply. The nature of this threat varies depending on the specific disease, crop, and farming conditions, making it challenging to conduct risk analysis and communicate

effectively with policymakers. Additionally, global changes such as climate change impact the types of disease threats faced, their potential outcomes, and the approaches for addressing them [1]. Farmers may struggle to diagnose plant diseases accurately due to their tiny features. Furthermore, many farmers lack the knowledge to diagnose diseases, so artificial intelligence can help them diagnose diseases more accurately [2].

Plant disease detection has seen a rise in the use of Machine Learning (ML) and Deep Learning (DL), which have shown promise for accurate disease detection from digital images [3]. Some commonly employed ML techniques include Support Vector Machine (SVM) [4–6], Artificial Neural Networks (ANNs) [7], K-means clustering [8], and Naive Bayes (NB) [9]. Due to their advances, DL techniques, especially CNNs, have gained popularity in computer vision. This advancement has prompted researchers to investigate using CNNs to detect plant diseases. Mohanty et al. [10] and Barbedo [11] initiated pioneering research in this direction. They utilized CNN models like AlexNet, VGG16, GoogleNet, and ResNet and employed the transfer learning technique within their approaches. In addition, researchers are working on using customized CNN models that incorporate convolution blocks from established CNN models like VGG16 and Inception Nets. These architectures include convolutional layers, pooling, and fully connected layers. The Inception module consists of a 3×3 max pooling layer and parallel convolutional layers with varying filter sizes. The outputs of these layers are then combined. The VGG model continues to serve as the foundation for numerous other architectures due to its straightforwardness and achieved second place in the ILSVRC 2014 competition [12–14].

Using a ViT has shown promise in developing models for detecting plant diseases. Researchers have investigated ViT to identify plant diseases on mobile devices and classify plant diseases automatically in real time. ViT-based models have performed excellently in classifying benchmark datasets such as CIFAR-100, Oxford-IIIT Pets, Oxford Flowers-102, and ImageNet. Proposed ViT-based models for plant disease detection include MobileViT specialized for plants (PMVT), Inception Convolutional Vision Transformer (ICVT), and hybrid approaches combining ViT with CNN [15–19].

The detection and effective management of diseases in crops are pivotal for sustaining agricultural productivity, ensuring food security, and fostering economic growth. Investments in disease detection and classification technologies protect farmers' livelihoods and contribute to the overall stability and prosperity of nations dependent on agriculture. Existing studies explain that the detection of plant diseases continues to be challenging, primarily because of

the numerous disease species and diverse crops involved. The similarity in symptoms between different diseases and the changing patterns of these diseases over time further compounds the complexity. The limitation is that traditional approaches have struggled to extract relevant features from input images. Consequently, researchers prefer deep learning models, especially CNNs, which can automatically extract relevant and informative features from the input data, leading to superior performance. However, CNN models are limited in their ability to analyze the relationships between distant pixels, as they only consider the correlation between spatially neighboring pixels within the receptive field defined by the filter size. Furthermore, while effective, existing CNN models often require large datasets and substantial computational resources, presenting challenges for real-time applications in resource-constrained environments. Researchers have recently explored the use of attention mechanisms to address the challenges associated with the CNN model's limited ability to capture relationships between distant pixels. Additionally, most existing studies have focused on individual CNN models, with a limited investigation into the potential benefits of ensemble modeling approaches for improved generalization of results. Consequently, we have created a hybrid framework that combines multiple DL architectures and vision transformers to extract profound features through a hybrid approach, as shown in Fig. 1. The key points of this paper can be stated as follows:

- A hybrid framework, including deep-learning CNNs and vision transformers, is suggested for detecting and classifying plant leaf diseases.
- The suggested end-to-end framework captures the most distinct features, allowing for precise detection and multi-class classification of plant leaf diseases.
- The suggested model has been extensively experimented with two different types of plant leaf disease datasets (Corn and Apple) and compared to the cutting-edge models using the same datasets.

The other sections of this research paper are as follows: Sect. 2 discusses recent related works on identifying plant disease. Section 3 introduces a comprehensive description of the materials and methods. Section 4 introduces the results and discussion. Section 5 presents the managerial implications, while the paper's conclusion is found in Sect. 6.

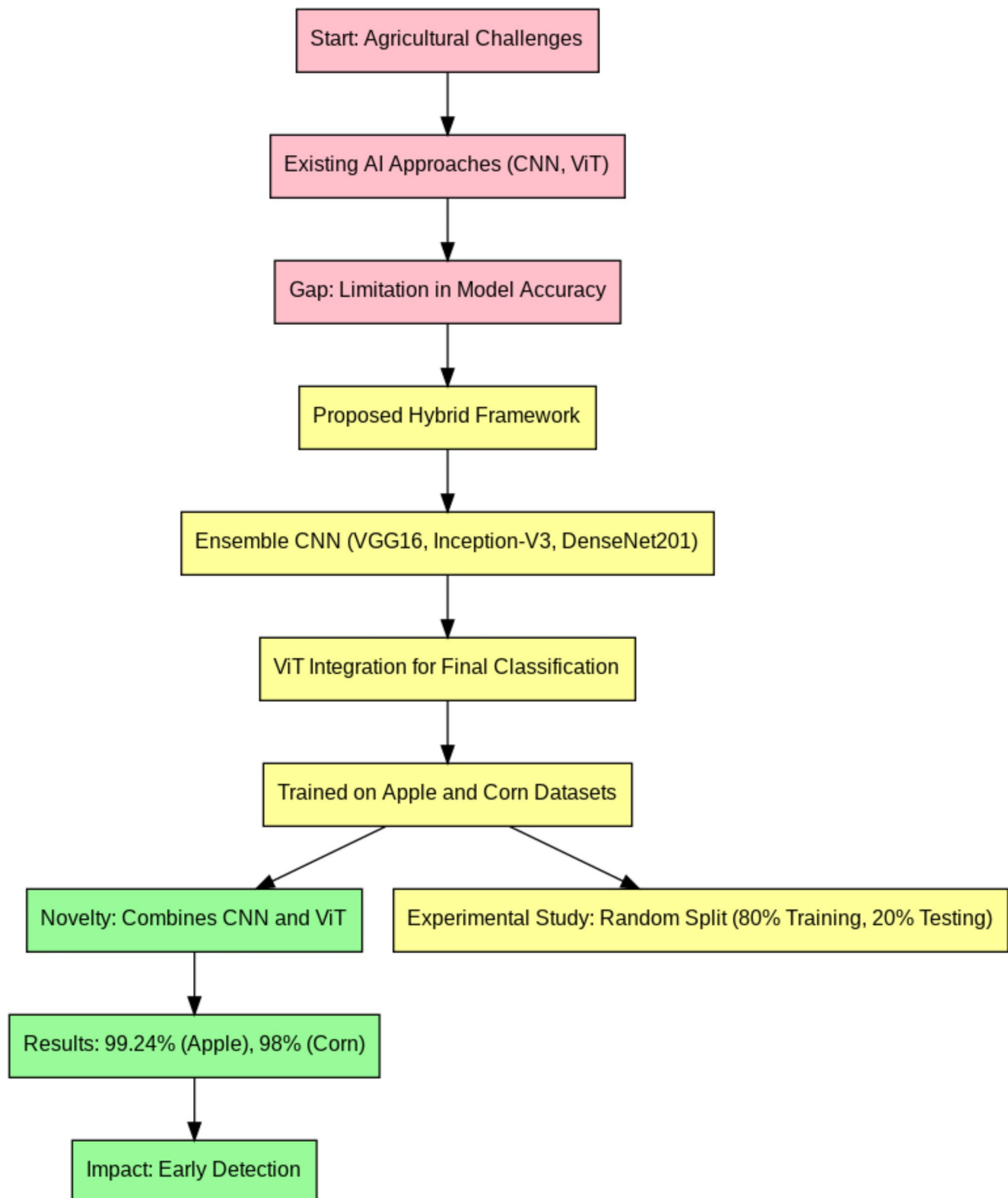


Fig. 1 Flow diagram of the proposed framework for accurately identifying plant diseases

Related works

Detecting plant diseases through machine learning and deep learning models has been a prominent area of recent research. Jiang et al. [20] developed a deep CNN architecture for detecting apple leaf disease. This architecture combined CNN with the Inception module to identify five types of apple leaf diseases. Bukumira et al. [21] proposed a Carrot grading system utilizing computer vision techniques and a cascaded graph CNN. This recognition framework demonstrated the capability to classify carrots accurately and grade them. Additionally, other studies have suggested DL models for identifying apple leaf diseases, including Fu et al. [22], Bansal et al. [23], and Khan et al. [24]. To enhance accuracy, these models employ diverse techniques like image augmentation, attention mechanisms, and dilated convolution. Studies have explored hybrid deep-learning models in various fields, such as agriculture and the medical sector, for disease diagnosis and classification [25–28]. These investigations have demonstrated the efficacy of combining deep learning architectures to effectively extract relevant features from the input data. Hosny et al. [29] developed a hybrid approach using lightweight deep model features and Local Binary Pattern features to detect and classify leaf diseases in apples, grapes, and tomatoes. This method resulted in accuracy rates of 97.8%, 97%, and 96% for each plant type, respectively. Priyadharshini et al. [30] created a CNN model by modifying the LeNet model to classify corn leaf diseases. A corn leaf image dataset from PlantVillage was used to train their approach. Their model demonstrated accuracy reached 97.89%. Waheed et al. [31] designed an optimized DenseNet structure to detect and classify corn leaf diseases: northern leaf blight, common rust, gray leaf spot, and healthy leaves. Their method resulted in an accuracy of 98.06%. Wu et al. [32] suggested a CNN model to detect Northern leaf blight on corn leaves using UAV images. They utilized the pre-trained ResNet architecture for feature extraction, employing its outcomes as inputs to a basic linear classifier. This model resulted in an overall accuracy of 95.1%. Researchers have also investigated the prediction and classification of plant diseases by employing a hybrid method that integrates CNNs with attention mechanisms. Zeng and Li [33] conducted experiments involving a residual CNN enhanced with self-attention to detect crop diseases. Their approach exhibited notable performance, achieving 98% accuracy on the MK-D2 dataset and a 95.33% accuracy rate on another dataset, AES-CD9214. Chen et al. [34] utilized a pre-trained MobileNet-V2 model, which had previously been trained using ImageNet, integrating an attention mechanism for identifying rice leaf disease, resulting in an accuracy of 98.48%. Qian et al. [35] introduced a novel model that diverges from conventional CNNs

by leveraging transformers and self-attention mechanisms. The model partitions visual data into small, localized segments, computes correlations between these regions using an attention mechanism, and then aggregates larger, global information to inform the classification task [35]. Reedha et al. [36] demonstrated the effectiveness of the convolutional-free Vision Transformer model, which leverages the self-attention mechanism to process an image as a sequence of patches through a standard transformer encoder. Despite using a relatively small dataset, they achieved high performance, which they attributed to techniques such as data augmentation, transfer learning, and a limited number of classes [36]. Wu et al. [37] employed a dual-scale approach by passing the dataset through two ViT models simultaneously, one with a small patch size and the other with a large patch size. The outputs of these two ViT models were then combined using a fusion model and fed into a multi-layer perceptron header. The researchers concluded that by integrating different scales of self-attention sequences, the model could extract richer information from the images at various levels of granularity [37]. The analysis of current methods utilized to identify plant diseases is summarized in Table 1. As shown in Table 1, the classification performance indicates a need for further improvements.

Materials and methods

The proposed architecture

The suggested hybrid DL architecture for plant leaf disease detection and classification is depicted in Fig. 2. The framework comprises three initial pre-trained CNNs: a VGG16 network, an Inception-v3 network, and a DenseNet 201 network, and it concludes with a ViT block [38]. The main contribution of the model is to combine pre-trained CNN architectures for effective extraction of deep features with a ViT that includes self-attention and Multilayer Perception (MLP) techniques for achieving accurate identification and classification results. CNN models can investigate the spatial relationship between adjacent pixels within a receiving area defined by the size of the convolutional filter while disregarding directional relationships and distance between these pixels [39]. However, recent advancements in transformers based on attention mechanisms are more effective and robust in considering the distance relations between pixels and their spatial correlation for improved accuracy in visual recognition operations. The suggested hybrid DL framework involves the following primary processing stages. Initially, the plant leaf images are resized to 128 by 128 pixels. Following this, data augmentation approaches are implemented to expand the dataset size, and

Table 1 Summary of related works for plant leaf disease detection and classification

Reference	Objective	Method	Key Findings	Limitations
Jiang et al. (2019) [20]	Real-time detection of apple leaf diseases	Improved CNN	- provide a high-performance solution with real-time detection	- Limited to apple leaf diseases. - The model's complexity may make it challenging to deploy on low-power devices.
Bukumira et al. (2022) [21]	Carrot grading based on computer vision and Graph Convolutional Neural Network (GCNN)	Cascaded Graph CNN with Bayesian optimization	- Effective in automated grading of carrots	- Specific to carrot grading, with limited adaptability for other crops. - The GCNN may be computationally intensive for real-time grading.
Fu et al. (2022) [22]	Lightweight model for apple leaf disease detection	Lightweight-CNN	- High accuracy with reduced model complexity	- Focused on apple leaves; the model might lack robustness for other crops.
Bansal et al. (2021) [23]	Apple leaf disease detection	ensemble of pre-trained deep CNN models	-Improved classification accuracy for apple diseases	- identification accuracy degrades for multiple diseases
Khan et al. (2022) [24]	Real-time apple leaf disease detection system	Lightweight-CNN and Yolov4	- Lightweight model with Real-time performance	- Classification performance needs improvement
Taji et al. (2024) [27]	Plant disease classification with hybrid CNN features	Metaheuristic-based hybrid CNN framework	- Enhanced performance with metaheuristic optimization	- Metaheuristic tuning is complex, which may lead to high computational costs and time-consuming parameter optimization.
Hosny et al. (2023) [29]	Multi-class plant leaf disease classification	Feature fusion of CNN and Local Binary Pattern	-Enhanced multi-class classification accuracy	- Complexity in feature fusion may impede model interpretability and adaptability.
Priyadharshini et al. (2019) [30]	Classification of maize leaf diseases	modified LeNet	-Lightweight model with Real-time performance	- Only evaluated for maize leaf disease classification
Waheed et al. (2020) [31]	corn leaf disease recognition and classification	- Optimized DenseNet model	- Improved classification efficiency and accuracy	- The dense architectural structure may elevate memory consumption, potentially impacting the deployment on devices with limited resources.
Wu et al. (2019) [32]	Autonomous detection of plant disease	CNN model on aerial imagery	- Effective in large-scale disease detection	- Limited to aerial imagery, which may miss finer disease symptoms.
Zeng & Li (2020) [33]	Crop leaf disease recognition	CNN model with Self-Attention	- Improved accuracy with self-attention mechanism	- Limited evaluations on crop datasets.
Chen et al. (2021) [34]	Rice plant disease detection	Pre-trained MobileNet-V2 with Attention Mechanism	- Efficient and accurate for rice diseases	- Lightweight structure may sacrifice detail on diverse or high-resolution datasets.
Qian et al. (2022) [35]	Maize leaf disease identification	Attention mechanism-based CNN	- Enhanced accuracy with self-attention	- High computational cost due to attention layers may limit real-time application.
Reedha et al. (2022) [36]	Weed and crop classification	attention-based deep network	- high performance, especially with small training datasets	- High-resolution image processing is computationally demanding, which may hinder real-time use.
Wu et al. (2021) [37]	Tomato leaf disease recognition	ViT	- multi-granularity feature extraction model	- High computational complexity of Vision Transformers may limit real-time use

subsequently, the data is partitioned into training, validation, & testing subsets. Second, the latest ensemble deep CNN structures (VGG16, Inception-v3, and DenseNet 201) extract DL features. Finally, the ViT receives these combined deep features of the CNN-based architectures. Here, the self-attention network identifies the varied symptoms within the images provided. The performance of detecting inaccurate symptoms in the dataset is then enhanced by using the MLP block. The ViT employing Multi-Head Self-Attention (MHA) offers an efficient approach for processing image patches and facilitates extracting prominent features within the patches. Combining CNN and the ViT network creates a potent feature extractor that integrates both local

and global features, thereby enhancing the interpretability of the model.

1.1. Pre-trained deep learning architectures

The following subsections discuss the pre-trained DL architectures used as the foundation of the suggested hybrid framework: VGG16 [40], GoogleNet [41], and DenseNet201 [42].

1.1.1. VGG16

The VGG16 convolutional neural network, created by Simonyan and Zisserman [40], has undergone training

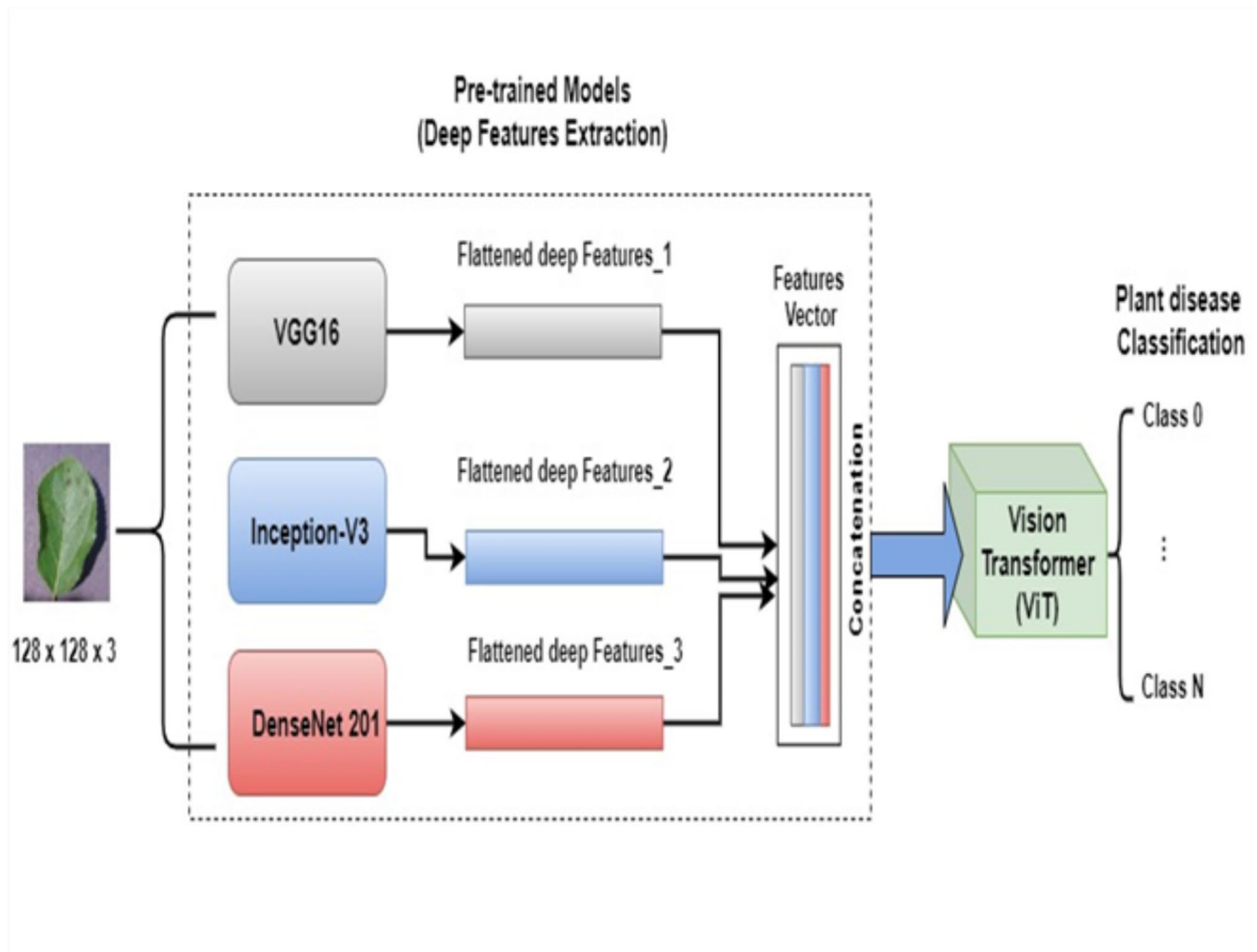


Fig. 2 Block diagram of the suggested model for plant leaf disease detection and classification

using the ImageNet dataset. We utilize a pre-trained VGG16 model that has captured important features from ImageNet's extensive dataset. We employ this VGG16 model and its pre-trained weights but exclude its original classification layers designed for different categories and integrate customized classification layers adapted to the new dataset. Subsequently, the weights of the convolutional layers are frozen to retain their learned capabilities from the initial training stage.

1.1.2. GoogleNet

The GoogleNet, also named Inception-V1, is a CNN architecture created by Google [41]. The GoogleNet model aims to incorporate filters of varying sizes that can operate simultaneously. Each inception module can capture different levels of significant features. For example, whereas the 5×5 conv layer extracts global features, the 3×3 conv layer is better at recognizing scattered features. The max-pooling layer focuses on extracting distinctive low-level features in

a local area. Subsequently, these diverse features are gathered and combined at a specific level before being transmitted to the next layer.

1.1.3. DenseNet201

DenseNet architecture facilitates information propagation between network layers by establishing connections from every layer to the following layers in a forward-propagating method while maintaining consistent feature map size. This is achieved by concatenating the previous layer's output with that of the subsequent layer. The transition layers consist of 1×1 convolutions and 2×2 average pooling. The Global Pooling (GP) layer is employed after the final dense block before applying SoftMax [42].

1.2. Vision Transformer (ViT)

As illustrated in Fig. 3, Dosovitskiy et al. [38] created the ViT architecture by adapting the original transformer encoder,

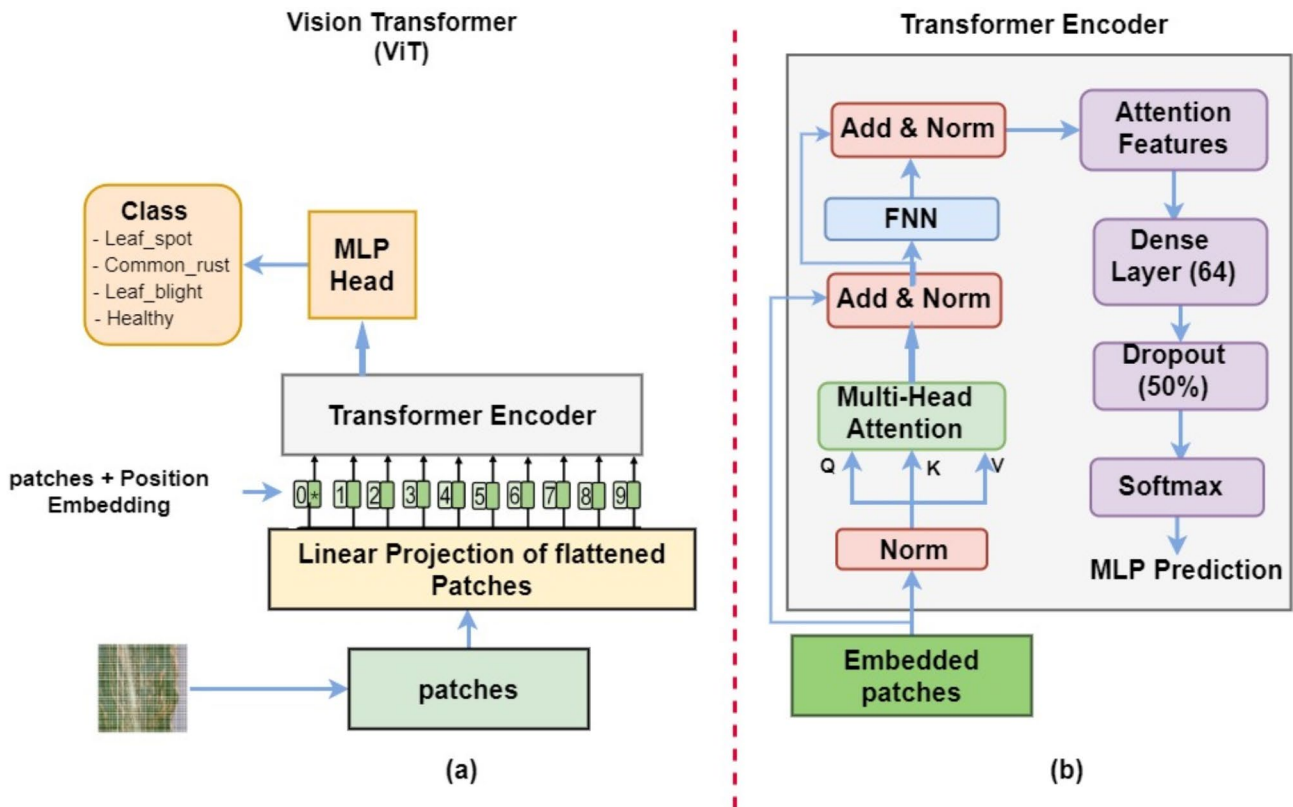


Fig. 3 Detailed description of ViT block (a) The ViT model introduced by Dosovitskiy et al. [38] (b) The transformer encoder architecture

which successfully addressed natural language processing challenges. The ViT comprises MHA blocks, networks of MLP with linear projection, and a positional embedding approach. To input an image into the ViT model, it undergoes splitting into non-overlapping patches of a fixed size. These patches are subsequently flattened and converted into forms that are lower in dimension. Each flattened patch experiences a trainable linear transformation to produce its respective linear projection and positional embedding [38]. The resulting vector from the linear projection and embedding undergoes further processing in a transformer block. This block encompasses MHA, position-wise Feed Forward Network (FFN), Normalization Layer (NL), and residual connections for both MHA and FFN. Each patch can be attended to individually through the MHA mechanism, thereby improving the model's capacity to extract long-distance relationships within the input data.

$$\text{Attention}(Q, K, V) = \text{SoftMax}\left(\frac{QK^T}{\sqrt{d}}\right) * V \quad (1)$$

The input embeddings are represented by the matrices Q (Query), K (Key), and V (Value), where d denotes the dimensionality of the K and Q vectors.

In the context of MHA, A is the input vector which transforms three distinct vectors $Q = AW_Q$, $K = AW_K$, and $V = AW_V$; wherein W_Q , W_K , and W_V represent the matrices of weight. The score matrix is produced by computing the dot product between Q and the transpose of K . Subsequently, the SoftMax activation function is applied to the output vector as detailed in Eq. 1 [15]. In addition, we include a Dense layer and use a Dropout layer before the SoftMax layer in the classification heads. These layers are customized explicitly for plant leaf disease classification.

1.3. Extraction and fusion of deep features

The research indicates that most advanced DL approaches for identifying plant leaf diseases depend on a single CNN. Ensemble learning has not been extensively explored in detecting and classifying plant diseases. This study utilizes feature concatenation as an approach to ensemble learning, aiming to capture robust, deep features. As depicted in Fig. 1, the foundational network of the suggested model is created with different pre-trained DL models like VGG16, Inception-V3, and DenseNet201. Combining features from various models allows the ensemble model to tap into a broader range of features [2, 43–45]. The concatenation of DL architectures comprising VGG16, GoogleNet, and

Table 2 Data distribution for the dataset of apple leaf disease

Class Index	Class Name	Training set size (80%)	Testing set size (20%)	Total
0	Apple_scab	800	200	1,000
1	Black_rot	800	200	1,000
2	Cedar_apple_rust	800	200	1,000
3	Healthy	1,316	329	1,645
Total		3,716	929	4,645

Table 3 Data distribution for the dataset of corn leaf diseases

Class Index	Class Name	Training set size (80%)	Testing set size (20%)	Total
0	Gray_leaf_spot	1,363	341	1,704
1	Common_rust	1,358	340	1,698
2	Northern_leaf_blight	1,363	341	1,704
3	Healthy	1,334	334	1,668
Total		5,418	1,356	6,774

DenseNet201 involved removing the classification layer from each model and extracting deep features from their last block convolutional layers. Upon removal of the classification layer, the VGG16 architecture produced an output of (None, 4, 4, 512), while the outputs for GoogleNet and DenseNet were (None, 2, 2, 2048) and (None, 4, 4, 1920), respectively. The distinct output from the GoogleNet architecture necessitated standardizing all output features, which

involved zero-padding the GoogleNet architecture before concatenating the features.

1.4. Datasets

Two publicly available datasets from PlantVillage [46] were utilized to evaluate the effectiveness of the suggested hybrid DL framework. The apple leaf dataset comprises 4,645 leaf images divided into four classes: Apple_scab, Black_rot, Cedar_apple_rust, and Healthy (see Table 2). Furthermore, we utilized the corn leaf dataset of 6,774 leaf images divided into four specific classes: Gray_leaf_spot, Common_rust, Northern_leaf_blight, and Healthy (see Table 3). As shown in Tables 1 and 2, the proposed architecture's experimental study utilizes a randomly split dataset, with 80% designated for training and 20% for testing. Furthermore, a validation subset is created from the training data, constituting 15% of the total dataset. Figure 4 depicts some leaf image samples taken from each dataset. Additionally, all plant leaf images are resized to 128×128 pixels.

1.5. Experiments setup

The proposed hybrid DL framework was developed using the TensorFlow platform and open-source Keras libraries. As indicated in Table 4, the training process utilized the optimizer Adam and the categorical cross-entropy loss function.

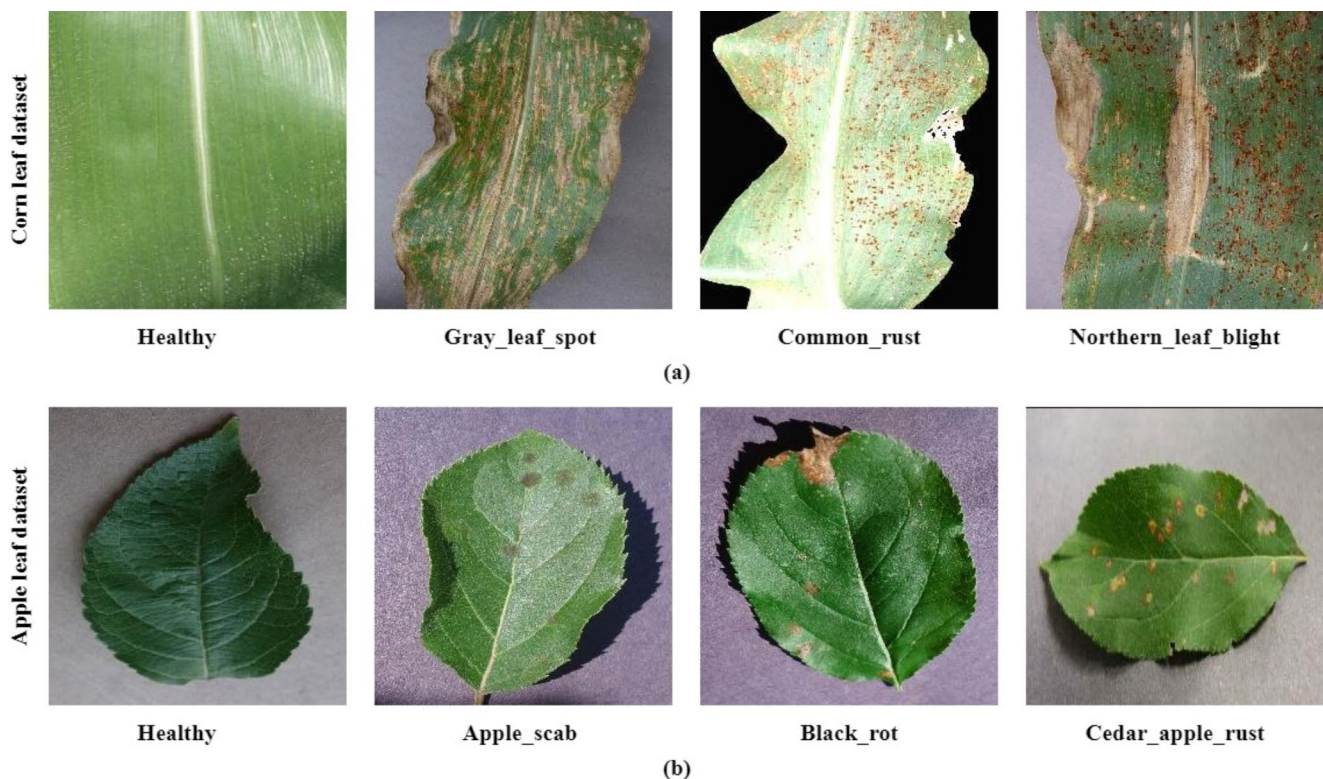


Fig. 4 Examples of the plant leaf dataset (PlantVillage [46]). (a) Corn leaf dataset, and (b) Apple leaf dataset

Table 4 Hyperparameters for the proposed framework

Function	Parameter	Value
Training Parameters	Optimizer	Adam
	Learning Rate	0.0001
	Epochs	50
	Batch Size	32
ViT Parameters	Batch Size	2
	Embedded Dimension	64
	Number of Attention Heads	8
	Number of MLP Layers	256

The experiment used a Learning Rate (LR) of 0.0001 with 50 epochs and incorporated the early stopping method with a patience of 10. Regarding the ViT, the patch size used was 2, with a dropout rate of 0.01 applied to all layers, and it utilized 8 heads. Additionally, an embedded dimension of 64 (signifying the dimension that allows high-dimensional vectors to be transformed to low-dimensional vectors without any loss) and the number of multi-linear perceptrons is 256. Additionally, all plant leaf images are resized to 128 by 128 pixels. These experimental procedures were carried out on the Google Colab platform to implement the suggested model.

1.6. Model evaluation

Standard performance measures, including accuracy, precision, recall, F1-score, the Area Under Curve Receiver Operating Characteristic (AUC-ROC) curve, and the Confusion Matrix (CM), are used to assess the suggested approach and each comparison model [47]. These assessment matrices are mathematically formulated as,

$$Accuracy = \frac{t_p + t_n}{t_p + f_p + f_n + t_n} \quad (2)$$

$$Precision = \frac{t_p}{t_p + f_p} \quad (3)$$

$$Recall = \frac{t_p}{t_p + f_n} \quad (4)$$

$$F1 \text{ score} = \frac{2t_p}{2t_p + f_p + f_n} \quad (5)$$

Here t_p , t_n , f_p , and f_n denote true positive, true negative, false positive, and false negative, respectively.

Table 5 Classification results of the models using unseen testing images of plant leaf diseases

DL Model	Evaluation metrics			
	Accu- racy (%)	Preci- sion (%)	Recall (%)	F1-score (%)
Apple	96	96	96	96
VGG16				
Inception-v3	94	94	94	94
DenseNet201	97	97	97	97
VGG16+Inception- V3+DenseNet201	97.6	98	98	98
Proposed Hybrid Model	99.24	99	99	99
Corn	96	96	96	96
VGG16				
Inception-v3	93.87	94	94	94
DenseNet201	97	97	97	97
VGG16+Inception- V3+DenseNet201	97	97	97	97
Proposed Hybrid Model	98	98	98	98

Results and discussion

This part focuses on presenting the outcomes of experiments conducted to assess the efficiency of the suggested hybrid architecture in detecting and classifying plant leaf diseases. It also includes a comparison of the performance of this model with other state-of-the-art CNN and ViT-based models. The effectiveness of various model architectures was assessed using the complete set of the provided datasets. First, individual assessment was conducted for each pre-trained model, like VGG16, Inception-V3, and DenseNet201. Then, the performance of a combination comprising VGG16, Inception-V3, and DenseNet201, and finally, the fusion of these pre-trained models and ViT were evaluated for detecting and classifying plant leaf diseases. As indicated in Table 5, the evaluation metrics utilized for comparative analysis include Accuracy, Precision, Recall, and F1-score, collectively providing a comprehensive assessment of the models' performance. The results for the Apple dataset indicate that the VGG16 model achieved a 96% performance across all evaluation metrics. While VGG16 demonstrated moderate performance, the Inception-v3 model performed slightly worse, reaching 94% across all metrics. In contrast, the DenseNet201 model outperformed both VGG16 and Inception-v3, achieving a 97% performance in all metrics, suggesting its superior capability in classifying apple leaf diseases. When the models (VGG16, Inception-v3, and DenseNet201) were combined, the performance improved, with an accuracy of 97.6%, precision and recall of 98%, and

an F1-score of 97%. Notably, the proposed hybrid model is outstanding compared to the other models, achieving an accuracy of 99.24% and precision, recall, and F1-scores of 99%. This indicates that combining CNNs and ViT, the hybrid approach provides significantly enhanced classification performance on Apple leaf diseases. The VGG16 model exhibits consistent performance for the corn dataset, achieving 96% across all evaluation metrics. In contrast, the Inception-v3 model demonstrates slightly inferior results, achieving an accuracy of 93.87% and 94% in precision, recall, and F1-score. Notably, the DenseNet201 model emerges as a robust individual model, reaching 97% in all metrics. Moreover, the ensemble model matches the performance of DenseNet201 alone, achieving 97% across all metrics. Also, the proposed hybrid model outperforms the other models, reaching 98% accuracy, precision, recall, and F1 score.

Figures 5 and 6 indicate that the suggested model's training was stopped after 26 and 27 epochs for apple and corn datasets, respectively. As shown in Fig. 5, the training accuracy rate reached 99.9%, and the validation accuracy reached 98.7% for the Apple dataset. In addition, Fig. 6 indicates

that the training and validation accuracy for the corn dataset reached 99.8% and 98.15%, respectively. The CM of the suggested hybrid model was employed for measuring the number of accurately and mistakenly detected samples for the apple and corn datasets, as depicted in Figs. 7 and 8. The CM is a valuable analytical tool for assessing the performance of a classification model, as it provides insights into the model's capability to accurately classify diverse types of plant leaf diseases. As shown in Fig. 7, the proposed hybrid model demonstrated high accuracy in classifying samples for Apple_scab, correctly identifying 196 out of 200 samples, with only 3 misclassified as Black_rot and 1 as Cedar_apple_rust. For Black_Rot, the model performed perfectly, correctly classifying all 200 samples. Similarly, for Cedar_apple_rust, the model correctly classified 199 out of 200 samples, with only 1 misclassified as Apple_scab. Finally, the model correctly identified 325 out of 329 samples for the Healthy class, with 2 misclassified as Apple_scab and 2 as Black_rot. For the Gray_leaf_spot class of the corn dataset (Fig. 8), the proposed hybrid model correctly identified 320 out of 341 samples, with 21 misclassified as Northern_leaf_blight. For Common_rust, the model correctly identified 337 out of 340 samples, with only 2 misclassified as Gray_leaf_spot and 1 as Northern_leaf_blight. For

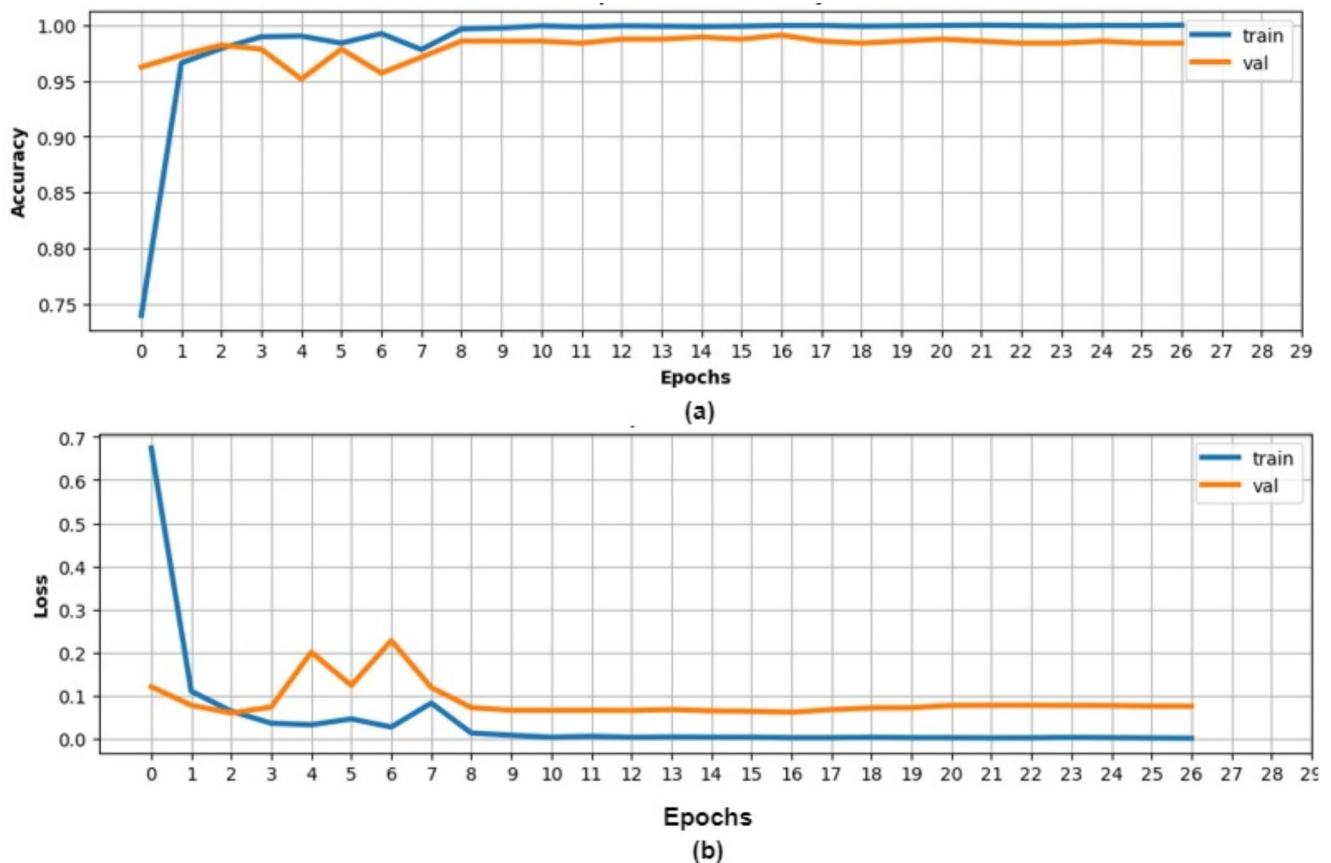


Fig. 5 The Apple dataset's accuracy and loss graph for the suggested model. (a) Training & validation accuracy, and (b) Training & validation loss

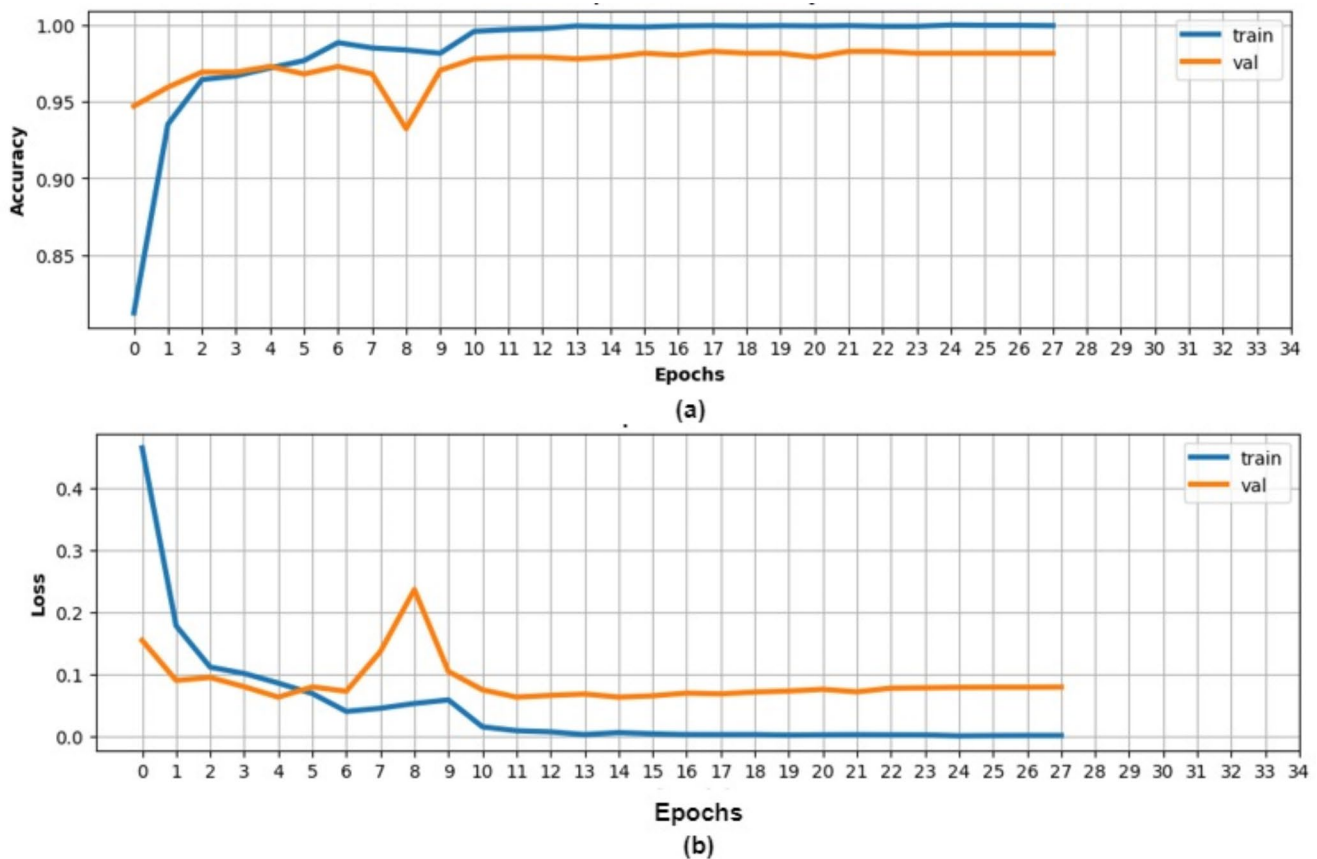


Fig. 6 The corn dataset's accuracy and loss graph for the suggested model. (a) Training & validation accuracy, and (b) Training & validation loss

Northern_leaf_blight, the model classified 334 out of 341 samples, with 7 misclassified as Gray_leaf_spot. Finally, the model performed perfectly for the Healthy class, correctly classifying all 334 samples. Generally, the CM demonstrates the robust performance of the hybrid model across the different classes for the Apple and Corn datasets. We assessed the AUC-ROC outcomes for individual classes within the apple and corn datasets to measure the effectiveness of the suggested model, as depicted in Figs. 9 and 10. The AUC-ROC curves for the apple and corn datasets indicate that the model excels in differentiating between various classes of plant diseases. The perfect AUC scores of 1.0 and 0.99 across all classes underscore the model's robustness and capability to identify plant diseases correctly. Table 6 compares the newly introduced plant disease classification models and the suggested hybrid DL model. This comparative study evaluates state-of-the-art models for the classification of plant diseases. The proposed model exhibits superior performance, achieving outstanding accuracy rates of 99.24% and 98% on the apple and corn datasets, respectively, outperforming all other models presented in the table. This remarkable success can be attributed to combining deep features extracted from three pre-trained CNN architectures

(VGG16, Inception-V3, and DenseNet201) and the incorporation of ViT technology.

Therefore, this hybrid framework combines the strength of CNNs and ViT to produce more robust features for plant leaf disease classification. The classification process's accuracy and efficiency are improved using an ensemble approach that includes three pre-trained architectures and ViT. The suggested model demonstrated outstanding accuracies of 99.24% and 98% for the apple and corn datasets, respectively, surpassing other cutting-edge models based on CNN and transformer architectures.

It is crucial to recognize the limitations of this technology. These limitations encompass the need for broader data diversity to ensure reliable generalization, enhanced interpretability to foster trust among end-users, and scalability to accommodate larger agricultural areas. Additionally, the high degree of similarity in disease patterns, colors, and textures, coupled with diseases that do not exhibit early symptoms on plant leaves, pose significant obstacles in developing effective AI-based solutions. Additionally, the emergence of new disease species and their altered spread behavior due to climate change further exacerbate the problem. Addressing these challenges necessitates the exploration and implementation of novel methods and techniques.

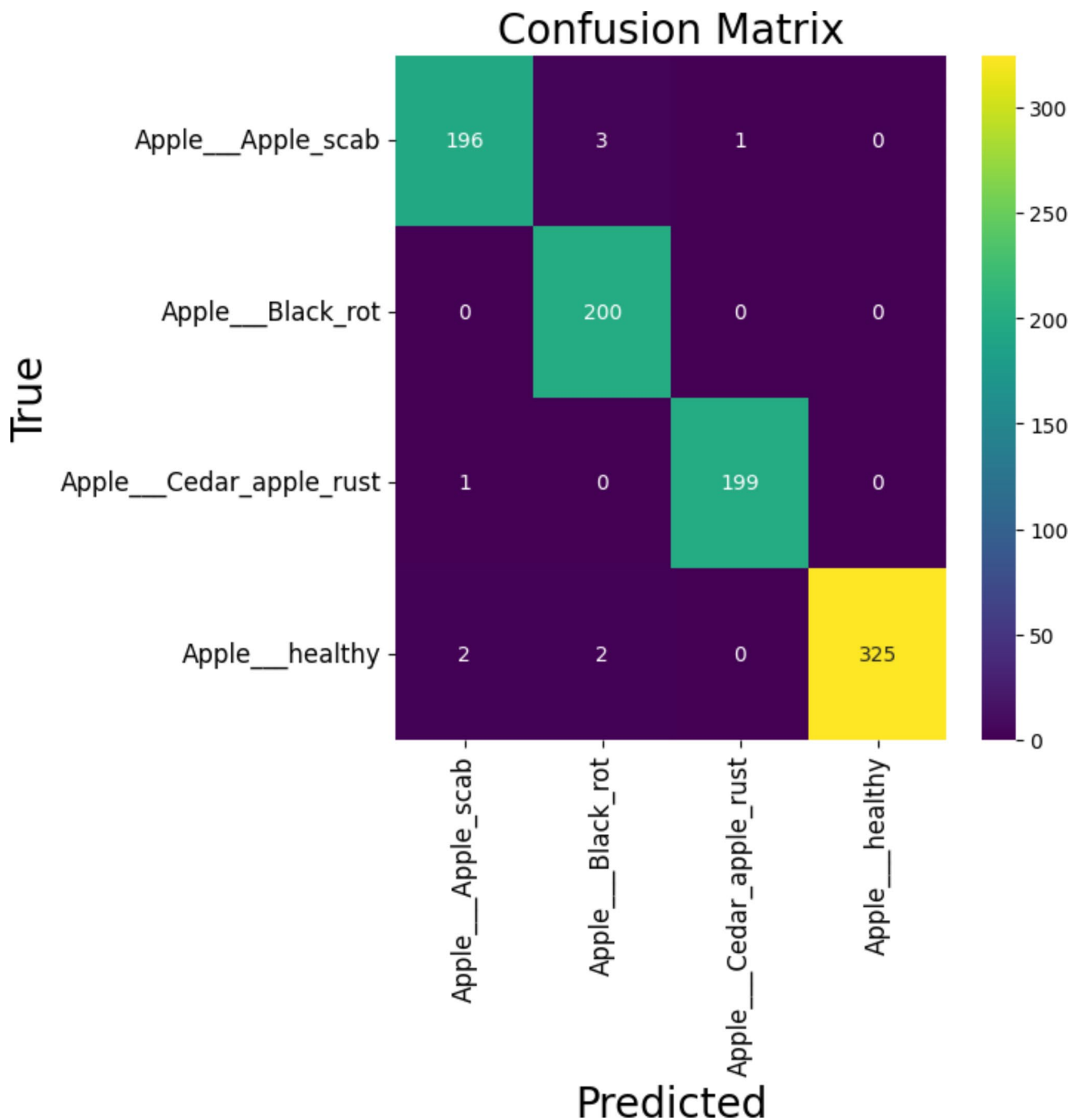


Fig. 7 The confusion matrix of the suggested hybrid model using the Apple dataset

Collaborating with subject matter experts, integrating with agricultural equipment, and addressing ethical data management are all crucial steps to further refine and enhance the system's capabilities. By embracing these challenges and pursuing these future directions, the agricultural sector can fully leverage the potential of DL-based systems, leading to sustainability in agriculture practices that contribute to economic prosperity.

Managerial implications

The proposed hybrid framework, which integrates CNNs and ViT, provides a solution for accurately and efficiently classifying plant leaf diseases in agricultural environments. The framework can achieve robust generalization across diverse plant species and disease types by combining CNNs for feature extraction and ViT for extracting global

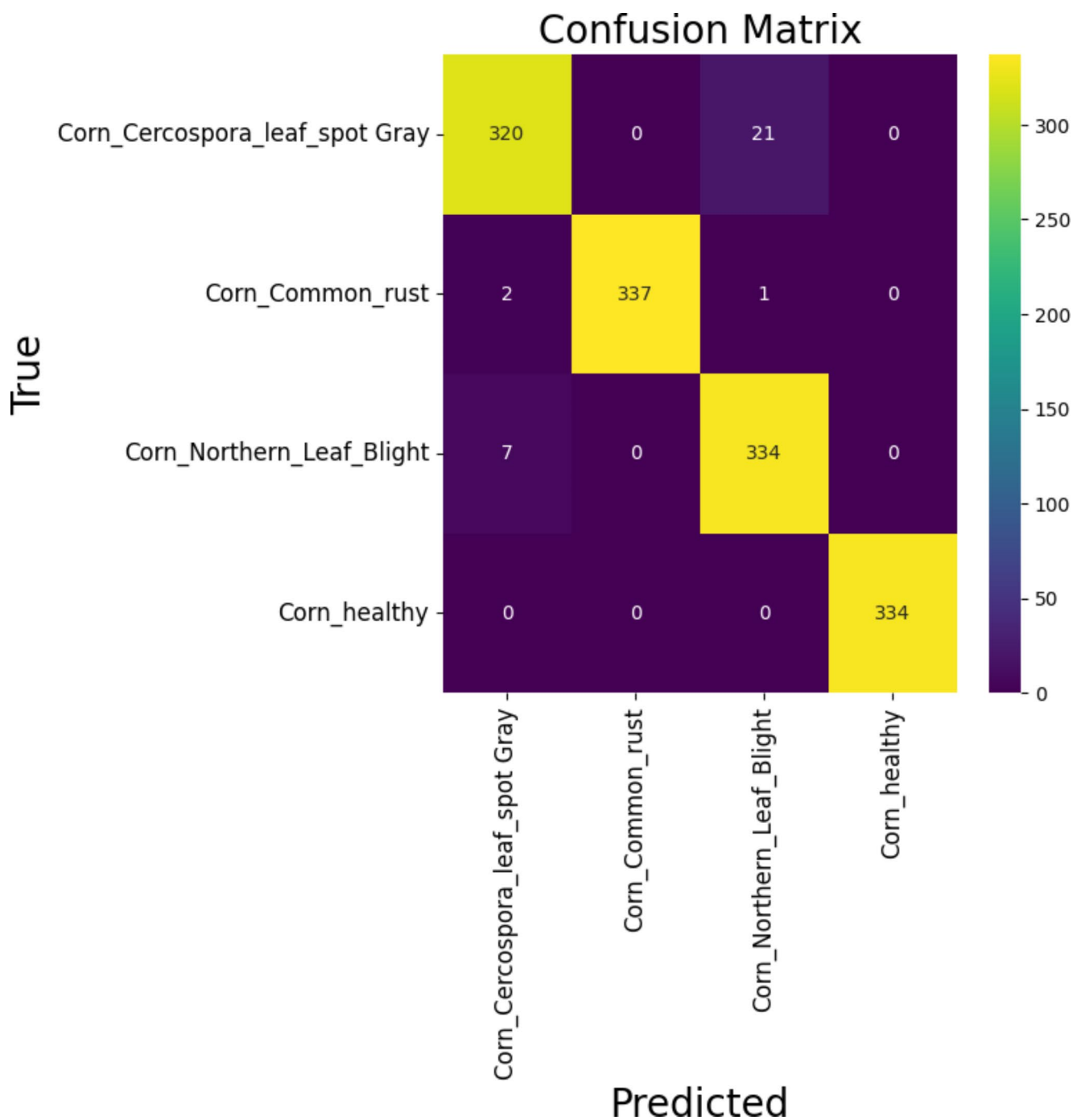


Fig. 8 The confusion matrix of the proposed hybrid model using the Corn dataset

interdependencies. This framework has been validated through comprehensive experiments on the Apple and Corn datasets, demonstrating its superior performance compared to cutting-edge methods.

Implementing this framework can substantially enhance the early identification of plant diseases in practical farming applications, bolstering precision agriculture. Additionally, using transfer learning and pre-trained models addresses

the challenge of data scarcity. This hybrid framework can be adopted to create automated plant disease identification systems, leading to improved crop management and yield optimization while mitigating the impact of plant diseases on food production. Furthermore, the success of this framework underscores opportunities for additional research and development, including its potential for scalability,

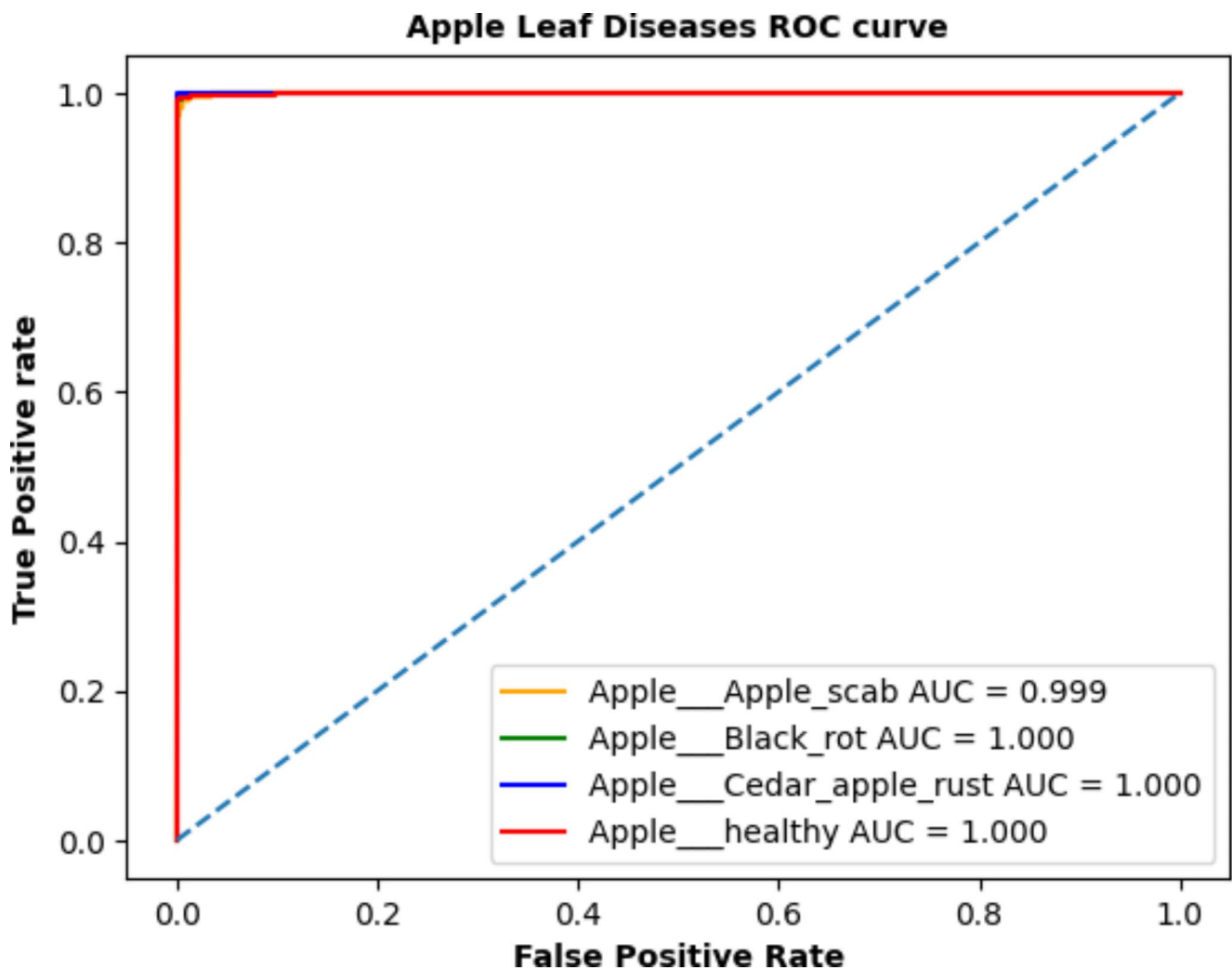


Fig. 9 The AUC-ROC results from the suggested hybrid model using the Apple dataset

real-time implementation, and adaptability to other crops and diseases.

Conclusion

This study introduces a hybrid framework combining CNNs and ViT to identify and classify plant leaf disease. Initially, three pre-trained CNN architectures (VGG16, Inception-V3, and DenseNet201) are employed to extract the features of the leaf. The weights of these pre-trained models are fine-tuned using the ImageNet dataset. Subsequently, the ViT is leveraged to extract the deep features of the leaves. Finally, the MLP head classifier determines the leaf's class. The proposed model's efficacy was assessed through accuracy, precision, F1-score, and recall metrics. Two publicly available datasets from the PlantVillage repository (apple and corn) were utilized to evaluate the proposed framework's effectiveness. The suggested hybrid model achieves remarkable

accuracies of 99.24% and 98% for apple and corn datasets, respectively. Additionally, the suggested model's performance for detecting and classifying plant leaf diseases surpasses numerous cutting-edge models. This model shows much potential for further evaluation using various plant datasets, which can provide valuable assistance to the industry in safeguarding livelihoods and offering intelligent service options to the farmers. The proposed model is accurate and applicable in various plant, fruit, and plant-stem datasets. Also, the proposed model is a potential tool for various computer vision applications and challenges such as masked face recognition, cancer detection and classifications, and image-based industrial quality control.

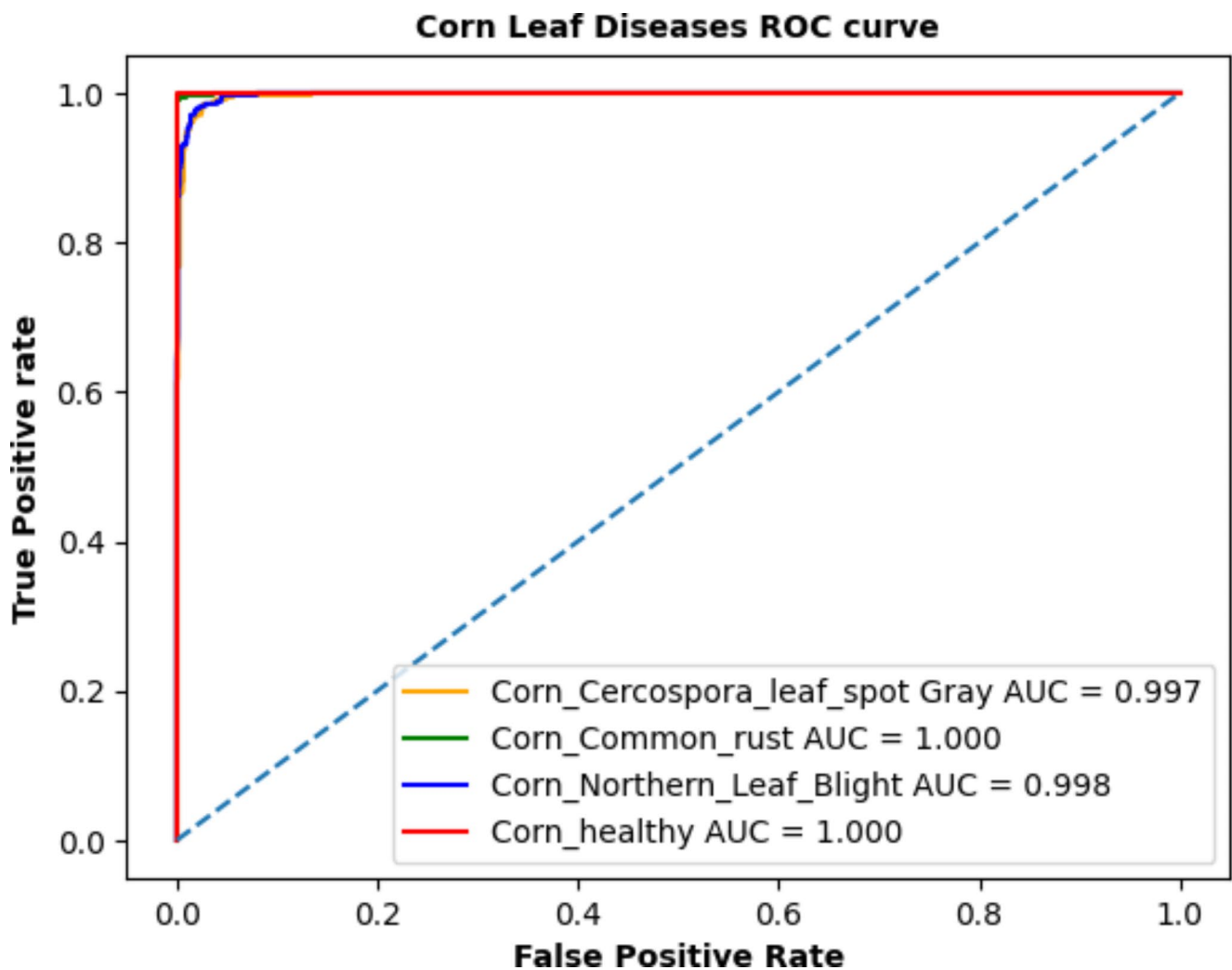


Fig. 10 The AUC-ROC results from the suggested hybrid model using the corn dataset

Table 6 Comparison with the other cutting-edge approaches for plant leaf disease (PlantVillage dataset [46])

Dataset	Authors	Approach	Accuracy (%)
Apple	Thakur et al. (2021) [48]	CNN+ViT	98.6
	Li et al. (2022) [49]	CNN+ViT	96.7
	Arshad et al. (2023) [44]	CNN+ViT	96.42
	The proposed hybrid model	An ensemble of pre-trained DL models with ViT	99.24
Corn	Waheed et al. (2020) [31]	optimized DenseNet model	98
	Qian et al. (2022) [35]	Attention-based and transformer	97.7
	Mishra et al. (2020) [50]	DCCN	88.46
	The proposed hybrid model	An ensemble of pre-trained DL models with ViT	98

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Data availability Data will be available upon request.

Declarations

The Conflict of Interests/Competing Interests No conflict of interests.

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